

Calligraphy Geometry and Topology: A Calligraphy Encryption Framework for Higher-Order Infinitesimal Entanglement and True Random Generation

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Abstract

This paper presents a complete mathematical framework for *Calligraphy Encryption*, which regards human handwritten calligraphy as a physical entropy source and generates cryptographically secure true random sequences through higher-order infinitesimal entanglement structures. The core of the framework lies in that the combinatorial generation space of the eight primitive functions of regular script under stroke-order rules possesses a derived tangent complex structure, whose higher cohomology groups carry incompressible physical entropy; running script and cursive script correspond to continuous deformations of differentiable manifolds and fiber bundles. As a randomness extractor, the F-field maps homotopy invariants of entanglement structures into cryptographic hash outputs, realizing a deep integration of humanistic artistic features and encryption security.

Keywords: calligraphy encryption; higher-order infinitesimal; derived geometry; true random number; F-field; operad; chaos amplification; physical oracle

1 Introduction

1.1 Problem Background

Current random number generation methods fall into two categories:

- *Pseudorandom numbers*: generated by deterministic algorithms, inherently predictable.
- *Physical random numbers*: rely on quantum noise, thermal noise, etc., lacking human interpretability.

As a traditional Eastern art, calligraphy integrates subjective will, motor control, and physical environmental disturbances during creation. Each real writing is naturally unique at the microstructural level, providing an ideal physical entropy source for true random number generation.

Higher-order infinitesimal entanglement structures are the mathematical origin of true random numbers.

Treating each stroke as an independent function, regular script contains eight primitive functions; stroke-order rules belong neither to group theory nor topology, but a rewriting grammar. Under these rules, the eight primitives generate new configuration spaces through five operations: concatenation, nesting, threading, overlay, and return. Differentiating this space yields higher-order infinitesimals in a 2^8 -dimensional entangled state. Running and cursive scripts require description by differentiable manifolds.

- Section 2: Eight Primitive Function System and Operad Structure
- Section 3: Configuration Space and Derived Tangent Complex
- Section 4: Higher-Order Infinitesimal Entanglement and Randomness Amplification
- Section 5: F-Field as Randomness Extractor
- Section 6: Differentiable Manifold Formulation of Running/Cursive Script

- Section 7: Rigorous Proof of True Randomness
- Section 8: Calligraphy-Chain Authentication System and Post-Quantum Security
- Section 9: Conclusion and Outlook

Let $\mathcal{B} = \{h, v, p, n, z, t, d, g\}$ be the set of eight primitives:

Symbol	Stroke	Mathematical Description
h	horizontal	$h : [0, 1] \rightarrow \mathbb{R}^2, h(t) = (t, 0)$
v	vertical	$v : [0, 1] \rightarrow \mathbb{R}^2, v(t) = (0, t)$
p	left-falling	$p : [0, 1] \rightarrow \mathbb{R}^2, p(t) = (t, -\sqrt{t})$
n	right-falling	$n : [0, 1] \rightarrow \mathbb{R}^2, n(t) = (t, t^2)$
z	fold	$z : [0, 1] \rightarrow \mathbb{R}^2$, piecewise linear
t	rising	$t : [0, 1] \rightarrow \mathbb{R}^2, t(t) = (t, \varepsilon t)$
d	dot	$d : [0, \tau] \rightarrow \mathbb{R}^2$, bounded impulse
g	hook	$g : [0, 1] \rightarrow \mathbb{R}^2$, terminal abrupt return

Each primitive is a smooth mapping with parameters; singularities are allowed at endpoints.

Stroke-order rules are a *Rewriting Grammar* or *Coloured Operad*:

$$\mathcal{R} : \mathcal{B}^* \rightarrow \mathcal{C}$$

where $\mathcal{B}^*$ is the free monoid of strokes, and $\mathcal{C}$ is the set of valid Chinese characters.

Definition 1.1 (Stroke Combination Operations). 1. *Concatenation*: $b_1 \cdot b_2$, endpoint connection

$$\text{Conn}(b_1, b_2) = b_1(1) = b_2(0)$$

2. *Nesting*: $b_1 \circ b_2$, one stroke enclosed by another

$$\text{Nest}(b_1, b_2) = b_2([0, 1]) \subset \text{Int}(\text{Hull}(b_1))$$

3. *Threading*: $b_1 \bowtie b_2$, crossing at interior points

$$\text{Thread}(b_1, b_2) = \exists! t_1, t_2 : b_1(t_1) = b_2(t_2), t_1, t_2 \notin \{0, 1\}$$

4. *Overlay*: $b_1 \pitchfork b_2$, partial overlap (ink superposition)

$$\text{Overlay}(b_1, b_2) = b_1(D_1) \cap b_2(D_2) \neq \emptyset, \mu(b_1 \cap b_2) > 0$$

where μ is the ink measure. Overlay introduces noncommutativity and time-order dependence.

5. *Return*: $b \mapsto b^*$, terminal self-similar folding

$$\text{Ret}(b) = \lim_{\varepsilon \rightarrow 0} b(1 - \varepsilon) = b(1), \dot{b}(1) = -\lambda \dot{b}(1^-)$$

Proposition 1.1 (Existence of Operad Structure). $(\mathcal{B}, \mathcal{R})$ forms a coloured operad with color set $\text{Col} = \{\text{start}, \text{end}, \text{cross}, \text{overlay}, \text{nest}\}$.

Proposition 1.2 (Noncommutativity of Overlay).

$$\text{Overlay}(b_1, b_2; t_1 < t_2) \neq \text{Overlay}(b_2, b_1; t_2 < t_1)$$

Ink distributions differ due to penetration order:

$$\mu_{\text{pre}}(x) = \mu_0(x) + \alpha \int_0^{t_1} K(x, b_1(\tau)) d\tau$$

$$\mu_{\text{post}}(x) = \mu_0(x) + \alpha \int_0^{t_2} K(x, b_2(\tau)) d\tau + \beta \int_{t_2}^{t_1} K(x, b_1(\tau)) d\tau$$

where K is the ink diffusion kernel, α, β are penetration coefficients.

Let $\mathcal{W}$ be a stroke sequence of length n . Each stroke b_i has:

- shape parameter $\theta_i \in \Theta_i$ (curvature, length, thickness)
- position $x_i \in \mathbb{R}^2$
- orientation $R_i \in SO(2)$
- timestamp $t_i \in \mathbb{R}_{\geq 0}$

Definition 1.2 (Calligraphy Configuration Space).

$$\mathcal{M}_{\mathcal{W}} = \prod_{i=1}^n (\Theta_i \times \mathbb{R}^2 \times SO(2) \times \mathbb{R}_{\geq 0}) / \sim_{\mathcal{R}}$$

where $\sim_{\mathcal{R}}$ is the equivalence relation from stroke-order rules.

Differentiate the one-parameter family $\mathcal{W}(s)$:

$$\frac{d\mathcal{W}}{ds} = \sum_{i=1}^n \left(\frac{\partial \mathcal{W}}{\partial \theta_i} \frac{d\theta_i}{ds} + \frac{\partial \mathcal{W}}{\partial x_i} \frac{dx_i}{ds} + \frac{\partial \mathcal{W}}{\partial R_i} \frac{dR_i}{ds} + \frac{\partial \mathcal{W}}{\partial t_i} \frac{dt_i}{ds} \right)$$

Definition 1.3 (Derived Tangent Complex).

$$\mathbb{T}_{\mathcal{W}} = H^0(\mathbb{T}_{\mathcal{W}}) \oplus H^{-1}(\mathbb{T}_{\mathcal{W}}) \oplus H^{-2}(\mathbb{T}_{\mathcal{W}}) \oplus \dots$$

where

- H^0 : classical tangent space (low-order deformation)
- $H^{-k}, k \geq 1$: k -th order infinitesimal deformation

Proposition 1.3 (Dimension Formula).

$$\dim H^0 = \sum_{i=1}^n \dim(\text{free parameters}) - \dim(\text{constraints})$$

$$\dim H^{-k} = \infty \quad \text{for } k \geq 1$$

Consider stroke perturbations:

$$\delta b_i = b_i + \varepsilon \cdot \xi_i + O(\varepsilon^2)$$

ξ_i satisfies the coupled variational equation:

$$\sum_{j=1}^8 \mathcal{L}_{ij} \xi_j + \sum_{j,k=1}^8 \mathcal{M}_{ijk} \xi_j \xi_k = 0$$

where $\mathcal{L}_{ij} = \partial^2 \mathcal{R} / \partial b_i \partial b_j$, $\mathcal{M}_{ijk} = \partial^3 \mathcal{R} / \partial b_i \partial b_j \partial b_k$.

Theorem 1.1 (Higher-Order Entanglement Structure).

$$\dim \ker(\mathcal{L} + \mathcal{M}) = 2^8 - \text{rank}(\mathcal{R}) + \dim(\text{Overlay constraints})$$

corresponding to a 2^8 -dimensional entangled higher-order infinitesimal state.

Definition 1.4 (Kolmogorov Complexity). For a writing instance f , $K(f)$ is the length of the shortest program describing f .

Theorem 1.2 (Calligraphic Incompressibility). For real writing f :

$$K(f) \geq \dim_{\mathbb{R}} \mathbb{T}_{\mathcal{W}} - C$$

where C is the description length of $\mathcal{R}$.

Higher-order infinitesimal entanglement forms a dynamical system:

$$\frac{d\xi_i}{dt} = \sum_{j=1}^8 \mathcal{L}_{ij}(\xi_1, \dots, \xi_8) \xi_j + \sum_{j,k=1}^8 \mathcal{M}_{ijk} \xi_j \xi_k + \eta_i(t)$$

where $\eta_i(t)$ is physical noise.

Definition 1.5 (Lyapunov Exponent Spectrum). $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_8$

	Script	Lyapunov Spectrum	Chaos Strength
Proposition 1.4 (Script Chaos Spectrum).	Regular	$\lambda_1 \approx 0$	weak
	Running	$\lambda_1 > 0, \sum_1^2 \lambda_i \approx 0$	moderate
	Cursive	$\sum_1^k \lambda_i > 0, k \geq 3$	strong

Theorem 1.3 (Quantum Chaos Limit). In strong chaos, the mixing time $t_{\text{mix}} \rightarrow 0$, and the effective density matrix approaches the maximally mixed state:

$$\rho_{\text{eff}} = \lim_{t \rightarrow \infty} \frac{1}{t} \int_0^t |\psi(\tau)\rangle \langle \psi(\tau)| d\tau \approx \frac{I}{2^n}$$

Definition 1.6 (Complex-Valued F-Field).

$$F(x, y) = F_R(x, y) + i F_I(x, y)$$

Component	Role	Randomness Source
F_R (real)	seeder	low-order structure
F_I (imaginary)	amplifier	higher-order entanglement, physical noise
$\mathcal{I}(F)$	extractor output	topological/statistical invariant
$\mathcal{H}$ (SHA3-512)	strong extractor	cryptographic amplification

Definition 1.7 (Encrypted True Random Sequence).

$$S_f = \mathcal{H}(\text{Encode}(\mathcal{I}(F)))$$

where $\mathcal{H}$ is SHA3-512, and S_f is a 512-bit cryptographically secure sequence.

Theorem 1.4 (Uniqueness). If $f_1 \neq f_2$, then $S_{f_1} \neq S_{f_2}$ almost surely.

Theorem 1.5 (Unpredictability). S_f cannot be predicted by any deterministic algorithm.

Theorem 1.6 (Min-Entropy Lower Bound).

$$H_\infty(S_f) \geq \sum_{k=0}^K \alpha_k d_k - o(1)$$

where α_k is entanglement efficiency, $d_k = \dim H^{-k}$. The system has an infinite entropy source.

Theorem 1.7 (Global Verifiability). Any third party can reproduce S_f using public operators.

Theorem 1.8 (Robust Stability). Let $\tilde{f} = f + \zeta$, $|\zeta| \leq \delta$. The hash distance between S_f and $S_{\tilde{f}}$ decays exponentially with δ .

In running/cursive script, primitives fuse into a global manifold:

$$\mathcal{C} : S^1 \rightarrow \mathbb{R}^2 \times \mathbb{R}_{\geq 0}$$

Definition 1.8 (Jet Bundle).

$$J^k(\mathbb{R}, \mathbb{R}^2) = \{(t, x, \dot{x}, \ddot{x}, \dots, x^{(k)})\}$$

Cursive trajectories are sections satisfying the energy-minimization principle:

$$\mathcal{S}[\mathcal{C}] = \int \left(\alpha |\dot{\mathcal{C}}|^2 + \beta |\ddot{\mathcal{C}}|^2 + \gamma \kappa^2 \right) dt$$

where κ is curvature.

Definition 1.9 (Calligraphy Fiber Bundle).

$$\pi : E \rightarrow M, \quad \pi^{-1}(p) = \text{brush state space at } p$$

- Base manifold M : stroke center curve $\mathcal{C}(t)$
- Fiber F : brush shape + ink distribution

“Flying white” corresponds to fiber singularities.

Real writing: $f = f^* + \xi$, where ξ is physical noise with infinite-dimensional entropy.

Theorem 1.9 (Chaotic Sensitivity of Complex Fourier Phase). For $f_1 = f + \xi_1$, $f_2 = f + \xi_2$, $\xi_1 \neq \xi_2$, then $\varphi_{f_1}(u, v) \neq \varphi_{f_2}(u, v)$ almost surely at high frequencies.

Theorem 1.10 (Unique Separability of Complex Wavelet Features). For any two distinct writings $f_1 \neq f_2$,

$$P(\Psi_{f_1} = \Psi_{f_2}) = 0$$

Theorem 1.11 (Uniqueness of F-Field Invariants). If $f_1 \neq f_2$, then $\mathcal{I}(F_{f_1}) \neq \mathcal{I}(F_{f_2})$ almost surely.

Theorem 1.12 (Existence of Encrypted Random Sequence). For any real writing f , S_f exists and is uniquely determined.

Theorem 1.13 (Min-Entropy Bound).

$$H_\infty(S_f) \geq 512 - o(1)$$

Definition 1.10 (Physical Oracle). Calligraphy system acts as a physical oracle:

- Input: writing intention
- Output: F-field invariant $\mathcal{I}(F)$
- Internal: higher-order entanglement + physical noise + time-order irreversibility

Under quantum computing, traditional authentication faces quantum collapse, static leakage, and bottlenecks. Calligraphy encryption offers:

- unstructured high-dimensional dynamic features (resistant to Shor’s algorithm)
- one-time live behavior (anti-replay)
- endogenous non-stored key
- super-exponential feature space ($> 2^{512}$)
- time-order irreversibility of overlay

Script	Mathematical Structure	Feature
Regular	Coloured Operad / Constrained Space	discrete, rigid, finite-dim
Running	Manifold / Jet Bundle	continuous, energy principle
Cursive	∞ -Category / Derived Geometry	higher homotopy, singularities

Conjecture 1.1 (Calligraphy Manifold). The space $\mathcal{S}$ of all calligraphic trajectories is an infinite-dimensional stratified manifold with singularities. There is a natural inclusion:

$$\mathcal{S}_{\text{Regular}} \hookrightarrow \mathcal{S}_{\text{Running}} \hookrightarrow \mathcal{S}_{\text{Cursive}} \hookrightarrow \mathcal{S}_\infty$$

where $\mathcal{S}_\infty$ is the derived infinitesimal stack.

Definition 1.11 (Random Functor).

$$\mathcal{R} : D(\text{Calli}) \rightarrow \mathbf{FinSet}_{\text{prob}}$$

maps writing instances to finite probability spaces.

Theorem 1.14 (Functorial Randomness). $\mathcal{R}$ is faithful iff entanglement efficiency $\alpha_k > 0$ for all k . True randomness of calligraphy is a necessary consequence of its categorical structure.

2 Conclusion

This paper establishes a true random generation mathematical model – *Calligraphy Encryption* – using human writing as a physical entropy source. The framework converts the natural uniqueness of artistic creation into cryptographically secure true random sequences, unifying humanistic art and encryption security.

Core contributions:

1. Eight-primitive operad structure formalizing strokes and rules.
2. Derived tangent complex revealing higher-order infinitesimal entanglement.
3. Chaos amplification mechanism for physical noise.
4. F-field extractor mapping homotopy invariants to secure outputs.
5. Calligraphy manifold conjecture unifying regular, running, and cursive scripts.

The system does not rely on pseudorandomness; its randomness comes from the physical non-replicability of real writing, providing a new humanistic physical entropy source for cryptography.

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